CogniSense: A Multimodal Digital Biomarker Pipeline Integrating Speech, Digital Clock-Drawing, and Oculomotor Signals Towards Preliminary Cognitive-Risk Screening

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***Abstract*—Early cognitive-risk screening remains difficult in low-resource settings because clinically labeled datasets are scarce and conventional assessments require specialist expertise or costly equipment. We present CogniSense, a multimodal digital biomarker pipeline for preliminary cognitive-risk screening on commodity mobile devices. It comprises three independently developed modules: (i) a multilingual speech-analysis pipeline based on the pretrained openai/whisper-small automatic speech-recognition model, which combines transcription with acoustic and lexical feature extraction to produce an experimental cognitive-risk score; (ii) a hybrid digital Clock Drawing Test combining OpenCV-based geometric analysis with a ResNet-18 classifier, achieving 97.93% image-classification accuracy on a held-out, non-clinical Roboflow dataset; and (iii) a camera-based oculomotor module that estimates clinically motivated eye-movement metrics through deterministic signal processing, since no public dataset pairs consumer-camera gaze sequences with cognitive-status labels. Each module is reported only against the evidence available for it, and we discuss what these results do and do not support. A roadmap for supervised multimodal fusion on clinically labeled data completes a transparent, honestly scoped Step-0 screening framework.**

***Index Terms*—cognitive screening, digital biomarkers, speech analysis, digital Clock Drawing Test, oculomotor tracking, multimodal fusion**

I. Introduction

Alzheimer's disease (AD) and related dementias affect more than 55 million people worldwide, with prevalence expected to rise as populations age [1]–[3]. Because neuropathological changes begin years before overt symptoms, early screening offers an opportunity for timely intervention. However, PET imaging, MRI, and cerebrospinal-fluid biomarkers remain expensive and inaccessible for routine primary care, particularly in low-resource settings [14]. General physicians are often the first point of contact for older adults but lack rapid, objective, low-cost screening tools; conventional instruments such as the MMSE and MoCA require trained administration and say little about underlying speech, visuospatial, and oculomotor processes.

We present CogniSense, a multimodal pipeline combining speech analysis, a hybrid digital Clock Drawing Test (CDT), and camera-based oculomotor assessment using only consumer hardware. Rather than claiming diagnostic capability, it is intended as a Step-0 digital triage system that identifies individuals who may benefit from further clinical evaluation. We distinguish implemented components from proposed developments: the speech module is an implemented transcription and feature-extraction pipeline without classifier performance claims, the CDT module's CNN component is evaluated on a held-out image dataset, and the oculomotor module is an implemented signal-processing pipeline without classifier performance claims.

A. Contributions

1) A multilingual speech-analysis pipeline based on pretrained Whisper automatic speech recognition, acoustic feature extraction, and lexical analysis for experimental cognitive-risk scoring.
2) A hybrid digital CDT combining rule-based geometric analysis with a ResNet-18 visual classifier and temporal drawing dynamics.
3) A hardware-free oculomotor pipeline that estimates clinically relevant gaze metrics from MediaPipe Face Mesh without dedicated eye-tracking equipment.

4) A discussion interpreting each module's results against the signal it was actually validated on, and a concise roadmap for clinically supervised multimodal fusion.

II. Related Work

A. Speech-Based Cognitive Screening

Speech has emerged as a promising non-invasive biomarker for cognitive decline. Previous studies have shown that linguistic and acoustic characteristics extracted from spontaneous speech can distinguish cognitively healthy individuals from those with Alzheimer's disease [4], [5]. Acoustic characteristics such as pause structure, pitch variation, vocal energy, and speech rate may provide language-independent information relevant to cognitive screening [9]. Nevertheless, access to clinically labelled speech recordings remains limited, restricting the development and validation of supervised multilingual systems [10]. CogniSense addresses this engineering challenge by implementing a pretrained Whisper-based transcription and speech-feature extraction pipeline, while reserving clinical validation for future work.

B. Digital Clock Drawing Test

The CDT is a widely accepted instrument for assessing visuospatial and executive function. Digital implementations additionally capture drawing dynamics such as stroke order, pauses, and speed, providing richer behavioural information than static scoring alone [11].

C. Oculomotor Biomarkers

Antisaccade errors, prolonged saccadic latency, fixation instability, and impaired smooth pursuit are consistently associated with cognitive impairment and AD [6]–[8]. Existing studies mostly rely on laboratory infrared eye trackers, which limits primary-care use, whereas recent computer-vision methods enable approximate gaze estimation with consumer cameras [13].

III. System Overview

CogniSense is a Step-0 screening system for standard smartphones or tablets with a microphone and front-facing camera. It is a digital triage aid, not a diagnostic tool, and produces referral recommendations for subsequent clinical evaluation. Its three independently developed modules (speech, CDT, and oculomotor) are each implemented as an independent FastAPI service and reported according to the evidence available for them; their processing pipelines are shown in Figs. 1–3. The proposed fusion framework is presented in Section IX.

IV. Module 1: Multilingual Speech-Pattern Screening

A. Recording and Transcription

The module accepts a user-recorded sample of continuous reading or description of a displayed paragraph or scene; no public speech dataset is needed at runtime. The user selects a language or requests automatic detection; the interface supports English, Hindi, Marathi, Bengali, Tamil, Telugu, Gujarati, Kannada, Malayalam, and Punjabi. The complete pipeline is shown in Fig. 1.

Transcription uses the Hugging Face Transformers ASR pipeline with the pretrained openai/whisper-small model [12], with openai/whisper-base or openai/whisper-tiny as fallbacks. The model is not fine-tuned on dementia or cognitive-status labels. Recordings are converted to mono, resampled to 16 kHz, mean-centred, trimmed to the detected speech region (with short padding so word onsets and offsets are retained), amplitude-normalized, and limited to 35 s; Whisper processes the audio in 15-s chunks. For an explicitly selected language, the corresponding language instruction is passed to Whisper; in automatic mode, the language is inferred from the ASR output or from script-based detection. Empty, numeric-only, low-information, and script-mismatched transcripts are flagged, and the system then returns a localized low-confidence message recommending a new recording in a quieter environment.

B. Acoustic and Lexical Feature Extraction

Acoustic analysis uses 30-ms frames with a 10-ms hop. Frame-level RMS energy delimits speech and silence, and a silent interval of at least 200 ms counts as a pause. The acoustic features are 13 cepstral coefficient means, spectral centroid, roll-off, and bandwidth, zero-crossing rate, mean and standard deviation of vocal energy, pause count, pause ratio, mean pause duration, total silence ratio, mean fundamental frequency (F0) with its standard deviation and range, and estimated speech rate. F0 is estimated by autocorrelation over 70–500 Hz; speech rate is approximated from energy peaks during active speech and is therefore an acoustic estimate, not a verified syllable count. Fields for jitter, shimmer, harmonic-to-noise ratio, and formants exist in the implementation but are not estimated or used.

Lexical analysis operates on the normalized, tokenized Whisper transcript and yields word count, unique-word count, type-token ratio (TTR), Guiraud's R, mean word length, filler-word count and ratio, repeated-bigram ratio, and hapax ratio. With U unique words among N total words,

$$\mathrm{TTR} = \frac{U}{N}, \qquad R = \frac{U}{\sqrt{N}}. \tag{1}$$

Fillers are identified with language-specific vocabularies, repeated bigrams are adjacent word pairs occurring more than once, and hapax words occur exactly once.

C. Experimental Risk Interpretation and Evaluation Status

The acoustic and lexical measurements are combined into an experimental speech-risk score, which is communicated through four engineering bands (Table I). Because the pipeline has not been validated against clinician-assigned cognitive-status labels, the score is a speech-pattern indicator and must not be read as a probability of dementia, AD, or mild cognitive impairment (MCI). The thresholds are engineering cut-points for communication and follow-up, not clinically calibrated ones. The frontend may also display a six-level global risk

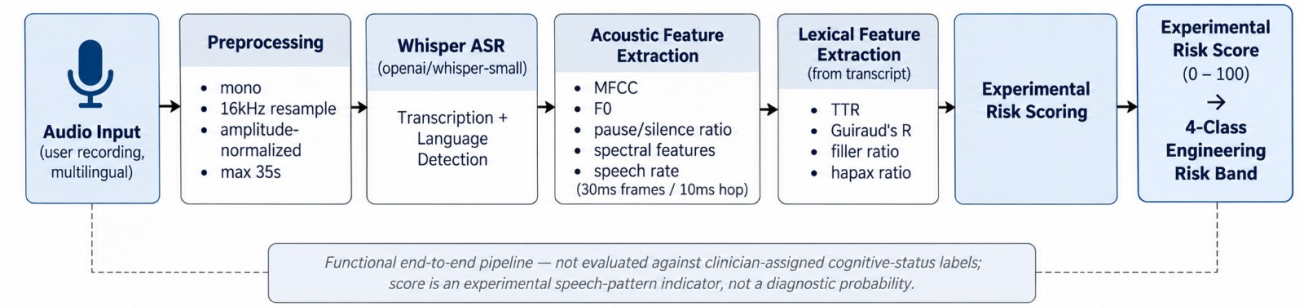


Fig. 1. Speech-analysis pipeline of Module 1: preprocessing, Whisper transcription, acoustic and lexical feature extraction, and experimental risk scoring. The pipeline is a functional end-to-end implementation and has not been evaluated against clinician-assigned cognitive-status labels.

scale, but the additional classes are presentation categories, not trained or validated speech classes.

TABLE I
RISK BANDS: SPEECH-MODULE ENGINEERING BANDS AND CDT FUSED-SCORE BANDS

Class	Interpretation	Speech score	CDT score
0	Cognitively Normal	0–24.9	0–24.9
1	Subjective Cognitive Decline	25–44.9	25–49.9
2	Mild Cognitive Impairment	45–64.9	50–69.9
3	High Risk, Urgent Referral	65–100	70–100

The module outputs the recording duration, transcript, detected or selected language, acoustic and lexical features, the risk score, an engineering risk class, and explanatory risk flags. It has been evaluated functionally as an end-to-end transcription and feature-extraction pipeline but not on clinician-labelled data; accordingly, no clinical accuracy, sensitivity, specificity, F1-score, or ROC-AUC is claimed. Its contribution is an interpretable multilingual workflow that can be evaluated on a future clinically labelled dataset.

V. MODULE 2: HYBRID DIGITAL CLOCK DRAWING TEST

A. Task Protocol and Scoring

Participants draw a clock showing a randomly assigned time on a touch canvas. Stroke-level pointer events yield temporal features (duration, stroke count, velocity, pauses, revisions). The exported image is analysed with OpenCV to estimate numeral completeness, hand geometry, ink distribution, quadrant coverage, centre usage, and structural consistency. A heuristic score $S_h \in [0, 100]$ from these geometric and temporal features is combined with the impaired-class probability of an ImageNet-pretrained ResNet-18 classifier (Fig. 2):

$$S_f = 0.65\, S_h + 0.35\, S_{c,100}, \tag{2}$$

where $S_{c,100}$ is the classifier probability scaled to 0–100. The fused score is mapped to four classes (Table I).

B. Results

The CNN was trained and evaluated on a binary-labelled Roboflow dataset with 725 held-out images (626 normal, 99 impaired); results are in Table II. The normal/impaired label is a pre-existing annotation supplied with the dataset, and we could not independently confirm the clinical rubric (e.g., Shulman- or Rouleau-style scoring) behind it; we therefore list this as an open provenance question rather than asserting a rubric we cannot verify.

TABLE II
CDT RESNET-18 HELD-OUT PERFORMANCE (IMPAIRED CLASS AS POSITIVE)

	Acc.	Bal. acc.	Prec.	Sens.	Spec.	F1	ROC-AUC
Value	97.93%	96.25%	91.18%	93.94%	98.56%	92.54%	0.9970

These results evaluate binary image classification only and are not clinical validation of the deployed four-class CDT score.

VI. MODULE 3: CAMERA-BASED OCULOMOTOR SCREENING

The module is a deterministic signal-processing pipeline rather than a supervised classifier, because no public dataset pairs consumer-camera eye movements with clinician-assigned cognitive labels (Fig. 3). It therefore reports extracted measurements only and makes no diagnostic performance claims.

Participants complete three standard paradigms: prosaccade (baseline saccadic behaviour), antisaccade (inhibitory control), and smooth pursuit (continuous tracking), which capture complementary executive and visuomotor functions associated with cognitive impairment. Gaze is estimated from MediaPipe Face Mesh iris landmarks using a front-facing camera at approximately 30 fps. From the gaze trajectory the system computes antisaccade error rate, saccadic reaction time, corrective-saccade latency, fixation stability, and smooth-pursuit gain, which correspond to established oculomotor

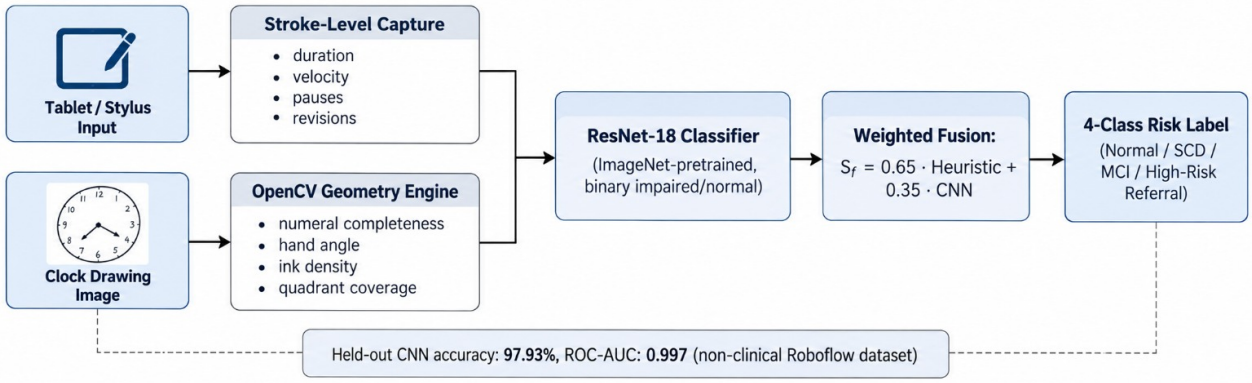


Fig. 2. Hybrid digital CDT pipeline of Module 2: stroke-level capture and OpenCV geometry analysis are combined with the ResNet-18 classifier through weighted fusion to produce a four-class risk label. The reported accuracy refers to the CNN component on a non-clinical Roboflow dataset.

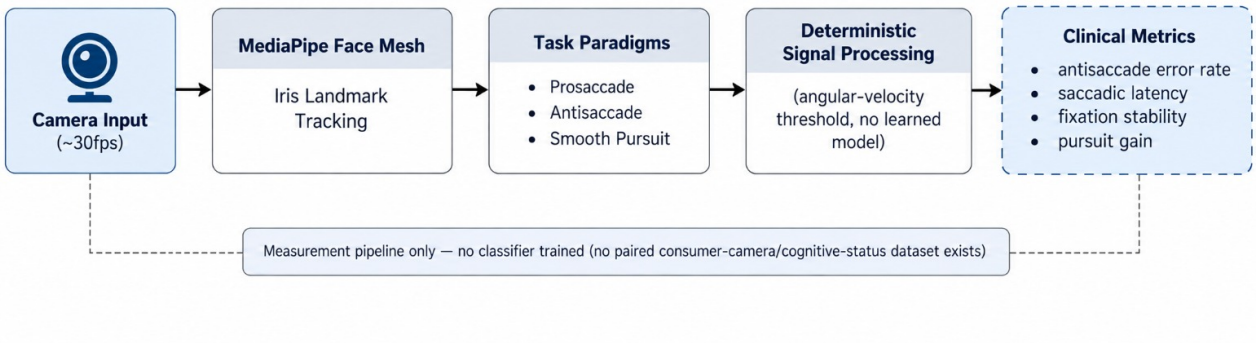


Fig. 3. Camera-based oculomotor pipeline of Module 3: MediaPipe Face Mesh iris tracking, three task paradigms, and deterministic signal processing yield clinically motivated gaze metrics. This is a measurement pipeline only; no classifier has been trained.

biomarkers, with the lower precision of consumer-camera tracking acknowledged. The acquisition and analysis pipeline is complete; no classifier has been trained or clinically validated, so no accuracy, sensitivity, or specificity is reported.

VII. Discussion

This section states plainly what each measured result does, and does not, support.

A. Speech Module

The module demonstrates the technical feasibility of combining pretrained Whisper transcription with explicit acoustic and lexical feature extraction in a multilingual screening workflow: Whisper turns a consumer-device recording into a transcript, while the signal-processing layer measures pause structure, pitch variation, energy, speech rate, and lexical diversity. These outputs demonstrate functional speech-biomarker extraction, not clinical discrimination between healthy participants and individuals with AD, MCI, or other disorders, since the pipeline has not been evaluated against clinician-assigned labels. Its clinical validity remains to be established on a purpose-collected, clinically labelled dataset.

B. CDT Module

The ResNet-18's 97.93% held-out accuracy (ROC-AUC 0.9970) shows that the Roboflow normal/impaired classes are highly separable by a standard CNN, consistent with the CDT's established sensitivity to visuospatial and executive deficits [11]. However, accuracy on this non-clinical corpus does not transfer automatically to the deployed fused four-class score (Table I), which also depends on the heuristic S_h and has not been evaluated against any ground truth. The result validates the CNN sub-component in isolation and motivates, but does not substitute for, end-to-end validation against clinically assigned CDT ratings.

C. Oculomotor Module

The absence of a classifier is a deliberate methodological choice, not an omission: we avoided training on a domain-mismatched proxy (e.g., free-viewing gaze datasets such as GazeCapture) merely to produce a reportable number.

The end-to-end extraction of clinically defined metrics from consumer-camera input via MediaPipe [13] demonstrates the technical feasibility of hardware-free oculomotor screening consistent with the laboratory literature [6]–[8], which we regard as a more defensible contribution than an inappropriately trained classifier.

D. Cross-Module Synthesis

Taken together, the three modules occupy different points on the same underlying problem identified by Petti et al.: restricted access to clinically labelled corpora is a central constraint in digital cognitive screening [10]. The speech module provides a functional Whisper-based biomarker extraction pipeline, the CDT module provides image-based analysis, and the oculomotor module provides deterministic gaze measurement without a supervised classifier. This difference in evaluation maturity motivates the clinical dataset and fusion roadmap described in Section IX. Each module runs as an independent FastAPI service deployable on a standard Android tablet or smartphone, supporting the intended Step-0, GP-facing deployment without specialized hardware.

VIII. Limitations

A. Speech Module

The module uses pretrained Whisper transcription and has not been fine-tuned or clinically validated for cognitive-status classification. Transcription errors may influence word count, lexical diversity, filler-word detection, and the risk score, and recording quality, noise, accent, speaking rate, and language may affect reliability. Although the interface supports multiple Indian languages, equivalent performance across them has not been established. The four risk bands are engineering thresholds not calibrated against MMSE, MoCA, clinical diagnosis, or specialist assessment. Clinical validation requires a purpose-collected dataset of recordings from cognitively normal participants, individuals with subjective cognitive decline, and clinically diagnosed participants, with standardized clinician-assigned labels.

B. CDT Module

The deployed score combines heuristic image analysis with a binary-trained CNN and is not clinically validated. Reported CNN performance rests on a public Roboflow dataset whose participant-level provenance and labelling rubric remain unverified (Section V).

C. Oculomotor Module

Consumer-camera iris tracking has substantially lower spatial and temporal precision than laboratory eye trackers; the accuracy and diagnostic utility of the computed metrics remain to be validated on clinically labelled data.

D. Ethical Considerations

CogniSense is exclusively a decision-support and preliminary screening system. It is not designed to diagnose AD, dementia, or MCI, replace clinical judgement, or guide treatment, and all outcomes should be interpreted alongside appropriate clinical assessment and external validation.

IX. Proposed Multimodal Fusion and Future Work

A. Fusion Strategy

Each module is to produce a calibrated probability of impairment (p_{speech}, p_{cdt}, p_{ocular}). Given the scarcity of paired multimodal data, weighted late fusion is preferred over transformer-based cross-attention for its sample efficiency, interpretability, and lower overfitting risk:

$$p_{\text{fused}} = \sigma(w_0 + w_1 p_{\text{speech}} + w_2 p_{\text{cdt}} + w_3 p_{\text{ocular}} + w_4 \left(p_{\text{speech}} \cdot p_{\text{cdt}}\right)), \tag{3}$$

where $\sigma(\cdot)$ is the sigmoid function. Cross-attention fusion remains a pre-registered comparison, to be benchmarked once paired data of sufficient scale (approximately 2,000 subjects) are available.

B. Clinical Dataset Roadmap and Targets

Clinical validation requires clinician-assigned cognitive labels. Planned datasets are: **Speech** – the DementiaBank Pitt Corpus with ADReSS/ADReSSo [15] for English-language training and benchmarking, plus a parallel Hindi/Marathi collection for cross-lingual work; **CDT** – clinically annotated CDT images from institutional collaborations, with confirmed rubric and participant-level splits; **Oculomotor** – a purpose-collected cohort (approximately 150–250 participants) performing the speech, CDT, and gaze tasks on the same consumer hardware under IRB approval and informed consent, which also supplies the paired data for fusion. Table III lists development goals to be tested, not obtained results; we would report a negative or null finding with the same transparency.

TABLE III
Proposed evaluation targets for the clinically supervised, fused system

Component	Target metric	Expected range
Speech (clinically validated)	Binary accuracy	To be established
CDT (clinically retrained)	Binary accuracy	85–93%
Oculomotor (pilot-trained)	Binary accuracy	70–80%
Multimodal fusion	Binary accuracy	82–90%

The modules probe largely distinct systems – word retrieval and processing speed (speech), visuospatial and executive planning (CDT), and frontal-executive inhibitory control (oculomotor) [8] – so fusion is expected, though not yet shown, to improve sensitivity to early impairment and reduce modality-specific false positives (e.g., ambient noise, motor tremor).

C. Remaining Future Work

1) Clinically validate the Whisper-based speech pipeline using a purpose-collected, clinically labelled speech dataset, with DementiaBank/ADReSS used only as an optional English-language benchmark and a separate Hindi/Marathi clinical collection used for regional-language validation.
2) Confirm CDT dataset provenance and labelling rubric; enforce participant-level splits.
3) Execute the paired 150–250-participant pilot under IRB approval.
4) Train and validate the late-fusion model (3), benchmark cross-attention against it, and add longitudinal monitoring and an explainability layer (e.g., SHAP over fusion coefficients) for GP-facing reports.

X. Conclusion

CogniSense presents a transparent multimodal framework for preliminary cognitive-risk screening using speech, digital clock drawing, and camera-based oculomotor analysis on consumer hardware. The speech module is implemented around pretrained Whisper transcription and explicit acoustic and lexical feature extraction, while the CDT module provides image-based analysis and the oculomotor module is implemented as a deterministic measurement pipeline without unsupported performance claims. By separating measured results from a plan for clinically supervised fusion, the system provides a reproducible foundation for clinical validation.

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